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DIRECT FILE SHARING APPLICATION

Network Programming Project — Topic Identification

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Network Programming Project

ONE LOVE. ONE FUTURE.

1. Actors & Problem Definition

System Actors

- **File Publisher**: Makes selected local files available to others.
- **File Seeker**: Searches for files and downloads them from other users.

The same user can take both roles.

Problem Definition

- **Find shared files**: Help users locate machines that offer the files they need.
- **Transfer files directly**: Download from the machine holding the file, without intermediary file storage.
- **Keep listings accurate**: Update the catalog when files are withdrawn or downloads fail.

2. Core Use Cases

Account & File Publication

- **UC1 — Register & Log In**
  Create an account and sign in.
- **UC2 — Publish Files**
  List local files for others to find.
- **UC3 — Cancel Publication**
  Remove a file from the shared catalog.

File Search & Download

- **UC4 — Search by Name**
  Find files and view machines offering them.
- **UC5 — Download a File**
  Select a machine and download directly.
- **UC6 — Report a Failed Download**
  Report the failure so the catalog can be updated.

3. Inputs & Outputs Specification

Account & Catalog Operations

- **Account Creation & Login**:
  - *In:* Username, password credentials.
  - *Out:* Auth status (success/failure), session state.
- **File Publication**:
  - *In:* File name, metadata, host machine address.
  - *Out:* Publication confirmation; catalog index updated.
- **Cancel Publication**:
  - *In:* Target file identifier to revoke.
  - *Out:* Revocation status; file removed from catalog.

Search, Download & Maintenance

- **Search for Files**:
  - *In:* Target file name / keyword query.
  - *Out:* List of machine addresses hosting matching files.
- **Direct File Download**:
  - *In:* Selected machine address, target file name.
  - *Out:* File content streamed to local storage.
- **Report Indexing Error**:
  - *In:* Unsuccessful download signal (failed host/file).
  - *Out:* Error logged; catalog updated to purge entry.



THANK YOU !